STRobustNet: Efficient Change Detection via Spatial–Temporal Robust Representations in Remote Sensing

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***Abstract*— Vision transformers have achieved impressive performance in addressing spatial–temporal inconsistencies of change detection (CD) due to their ability to model long-range dependencies in bitemporal features. However, applying the self-attention operation directly to bitemporal features causes feature confusion and incurs high computational complexity. In this article, we propose a novel CD framework based on spatial–temporal robust representation (STRobustNet). To avoid directly applying self-attention to bitemporal features, we derive a collection of STRobustNets for land-use classes of interest. These representations are used to transform spatial–temporally inconsistent bitemporal features into consistent bitemporal classification representations, enhancing model efficiency and reducing feature confusion. Specifically, to fully perceive and integrate the varied appearances of the same class while minimizing deviation from the current input samples, we design a robust representation generation module (RRGModule), which utilizes the universal spatial–temporal context provided by the entire dataset and the specific spatial–temporal context from the current bitemporal images to generate robust representations for the interested land-use classes, improving robustness to spatial–temporal inconsistencies. Then, these representations are used to activate bitemporal features in different class channels, producing spatial–temporally consistent classification representations. The CD is ultimately performed using these consistent representations, effectively avoiding false detections caused by spatial–temporal inconsistencies. Experimental results demonstrate that STRobustNet achieves performance comparable to top-performing methods and offers the fastest inference speed among transformer-based methods. Code and pretrained models are available at https://github.com/DLUTTengYH/STRobustNet.**

***Index Terms*— Change detection (CD), feature confusion, spatial–temporal robust representation (STRobustNet), transformer.**

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I. Introduction

CHANGE detection (CD) in remote sensing aims to identify pixel-level changes in land-use classes within corresponding areas of a pair of bitemporal satellite images captured at different times [1], [2], [3]. CD plays a crucial role in practical applications, including disaster management [4], [5], [6], damage analysis [7], [8], [9], deforestation monitoring [10], [11], [12], and urban planning [13], [14], [15]. However, accurately and efficiently detecting changed areas in bitemporal images still has challenges. The challenges stem primarily from the spatial–temporal inconsistency of bitemporal images: 1) inconsistency in the appearances of the same object captured at different times, resulting in temporal inconsistency and 2) inconsistency in the features of the same land-use class in different spaces, leading to spatial inconsistency. These inconsistencies can cause the methods to mistake pseudochanges as genuine ones [16], [17].

Recently, the integration of transformers' long-range modeling capabilities to enhance spatial–temporal consistency has been widely recognized as effective in achieving robust CD. The fundamental frame work of these methods comprises extracting bitemporal features, aggregating spatial–temporal context in bitemporal features [18], deriving change-aware features from bitemporal features, and performing CD on change-aware features [19] (or aggregating context for the change-aware features [20] before CD). For example, methods such as Hybrid-TransCD [1], STANet [13], and BIT [19] encode bitemporal images into semantic tokens and utilize the self-attention mechanism to aggregate all tokens in both spatial and temporal dimensions. However, aggregating context globally using self-attention operations entails high computational complexity. More importantly, the self-attention operator aggregates similar features even those from different classes, causing feature confusion [21], [22]. This leads to the model confusing different land-use classes. For instance, Fig. 1(a) shows a scenario where the building inside the box shares visual similarity with its surrounding environment. Fig. 1(b) illustrates the features of different regions after aggregation through the self-attention operator in BIT [19]. It can be observed that the buildings within the boxes in feature T0-1 and T0-2 blend with the surrounding environment, leading to BIT [19] mistakenly classifying the areas within the boxes as changed regions in Fig. 1(c).

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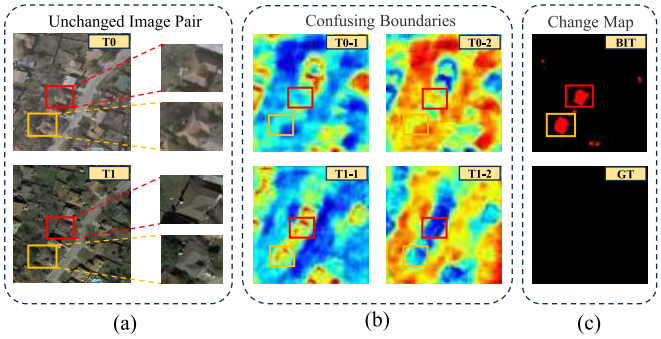


Fig. 1. Illustration of our motivation. (a) Bitemporal images. The boxes indicate some buildings exhibiting significant imaging differences across different times and similarities to the surroundings. (b) Distribution of different regions of interest indicated by BIT, which is derived through calculating the dot product between each semantic representation of BIT and the pixel-level features extracted from bitemporal images. In complex scenes, feature confusion makes BIT difficult to accurately highlight the buildings and the surrounding environment. (c) Change maps of BIT and ground truth. **Red** represents **false positives**, and **black** represents **true negatives**.

Therefore, how can we address the false detection caused by spatial–temporal inconsistencies without causing feature confusion and simultaneously maintain the method's efficiency?

In this article, we design a novel CD framework based on spatial–temporal robust representation (STRobustNet). To avoid feature confusion and ensure efficiency, we deviated from the previous approach of directly enhancing spatial–temporal consistency through global context aggregation on bitemporal features. Instead, we calculate a set of STRobustNets for land-use classes of interest, which unify the feature representations of the same land-use class across different times and regions. These representations are employed to activate bitemporal features in different class channels to obtain spatial–temporally consistent bitemporal classification representations. Finally, we perform CD based on these spatial–temporally consistent classification representations, successfully avoiding false detections caused by spatial–temporal inconsistencies.

In STRobustNet, the challenge lies in obtaining the STRobustNets. These representations should fully perceive and integrate different appearances of the same class while avoiding deviation from the input samples. Thus, we design a robust representation generation module (RRGModule). It is intuitively that in addition to the specific features of the input bitemporal images, the universal features provided by the entire dataset are more beneficial to the spatial–temporal robustness of the representations [23], [24]. Consequently, we introduce a group of learnable embeddings to capture the universal STRobustNets (STRR-U) of the land-use classes of interest. These learnable embeddings are adaptively updated during the training process, enhancing the model's ability to comprehend the diverse appearances exhibited by the same land-use class across the entire dataset. However, there may be variations in imaging conditions between different images, leading to deviations between the universal representations and the specific bitemporal images. Therefore, we iteratively refine these universal representations from the specific spatial–temporal features of the current input image pair. This operation aligns the universal representations with the specific bitemporal images, resulting in more spatially and temporally consistent classification representations.

Our STRobustNet is both concise and efficient. On the one hand, we obtain spatial–temporally consistent classification representations for bitemporal images using robust representations and determine whether changes have occurred based on these classification representations, thus solving the false detection problem caused by spatial–temporal inconsistency. On the other hand, by introducing STRobustNets, we avoid computing self-attention across the entire bitemporal features, thereby preventing feature confusion and significantly reducing computational complexity.

Our contributions can be summarized as follows.

1) We propose STRobustNet, a novel CD framework that detects changes by comparing the classifications of corresponding pixels calculated from STRobustNets of the interested land-use classes, effectively avoiding the issue of feature confusion while maintaining high efficiency.
2) We meticulously design the extraction method for robust representations, i.e., RRGModule, which utilizes the universal spatial–temporal context provided by the entire dataset and the specific spatial–temporal context from the current bitemporal images to generate robust representations for the interested land-use classes, enhancing the method's robustness to spatial–temporal inconsistencies.
3) We conducted comprehensive experiments on various datasets, demonstrating that STRobustNet obtains performance comparable to state-of-the-art methods and achieves the fastest inference speed compared to transformer-based methods.

II. Related Work

A. CNN-Based CD Methods

The current mainstream CNN-based approaches are feature-level methods, which typically extract bitemporal features from bitemporal images using the weight-shared Siamese network, as exemplified by SemiCDNet [25] and SRCD-Net [26]. UNet++ [27], FDCNN [28], and FC-Siam-conc [29] improve the ability to capture context information by stacking deeper neural networks. IFN [30], DTCDSCN [31], and SNUNet-CD [32] use the channel and spatial attention mechanisms to associate context information and extract spatial-robust features. In addition, to enhance the receptive field, methods involving cross-scale feature interaction and fusion have been investigated. These methods aim to supply both low-level features, which aid in target localization, and high-level features rich in semantic information. For example, Changer [33] exchanges bitemporal features in multiscale features to simultaneously perceive spatial–temporal context. Methods in [34] and [35] utilize an attention operation to recalibrate and integrate the multiscale features for feature enhancement. GateFormer [36] integrates the UNet with attention operation, proposing the gate attention U-shaped network to better leverage information from multiscale

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features. Despite achieving certain achievements, these methods still lack robustness to spatial–temporal differences due to the limitations of CNNs in modeling long-range dependencies and global contextual interactions. Recently, graph convolutional networks (GCNs) have been applied for feature extraction [37], [38], [39] to capture the local and global dependencies among the features of multitemporal images. However, the utilization of graph convolution significantly reduces the efficiency and flexibility of the method.

To effectively capture both local and global spatial–temporal contexts to enhance robustness against spatial–temporal inconsistencies, contemporary CNN-based approaches tend to incorporate the transformer operator within the CNN framework. These approaches leverage transformers to enhance feature interactions at the channel, spatial, and temporal levels, thereby enriching the local features of CNNs with global spatial–temporal context [18], [40]. A detailed exploration of these methods will be provided in Section II-B.

B. Transformer-Based CD Methods

With the widespread use of transformer [41], the accuracy of CD has been pushed to a new level. Some methods utilize transformers as auxiliaries to introduce global information into CNN features. For instance, ICIF-Net [18] fuses global features from transformer and local features from CNN at multiple scales to provide spatial–temporally robust features for CD. DMINet [40] employs cross-attention to aggregate bitemporal CNN features spatially and temporally. HANet [20] utilizes self-attention to aggregate spatial context for change-aware features. These methods benefit from transformers and exhibit improved performance. However, their application of self-attention or cross-attention, which involves computing attention between the entire feature maps, significantly increases model complexity. MLDANets [42] adaptively select and aggregate information from a few points across the global range of change-aware features, significantly reducing computational costs. Unlike MLDANets, in this article, we propose STRobustNet, which aggregates spatial–temporal context of the bitemporal features by our generated robust representations with much fewer numbers than the pixels, thereby greatly reducing model complexity.

Other methods primarily rely on transformers. Methods such as Changeformer [43], EGCTNet [44], SwinSUNet [45], and MSTDSNet-CD [46] use transformer to process each temporal feature separately and then input the processed features into the prediction head after concatenating or subtracting. MaskCD [47] revisits the CD task from the mask prediction and classification perspective, alleviating background noises and losses of details. However, the process of acquiring change-aware features involves the use of a Swin transformer and deformable multihead self-attention, which introduces complex computations. STANet [13], Hybrid-TransCD [1], and VcT [2] transform bitemporal images into semantic tokens and then utilize the self-attention mechanism to aggregate tokens in both spatial and temporal domains. This operation can enhance the spatial–temporal consistency. However, when features of different land-use classes exhibit similarities, the self-attention mechanism assigns higher weights to similar features during the aggregation process, which can lead to feature confusion between different classes. In this article, we explore a novel framework via STRobustNets for CD (STRobustNet). Our method can avoid the self-attention-based fusion of all semantic tokens, thus effectively mitigating the issue of feature confusion.

C. Methods Based on Other Technologies

There are also CD methods for remote sensing images that incorporate popular techniques such as GAN [48], Diffusion [49], CLIP [50], and SAM [51]. ChangeCLIP [52] enhances CD performance by integrating CLIP, leveraging the multimodal information of image–text pairs. Methods in [53] and [54] introduce the diffusion model into the CD task, enhancing the robustness to noise and accuracy at object edges in CD methods. The methods in [55] design a graph representation learning-guided diffusion model, combining the strengths of the diffusion model and GCN to improve the accuracy of CD. Although these methods achieve enhancements through new technologies, they also introduce significant computational complexity.

Some other methods utilize these new technologies to address misalignment and data scarcity. The methods in [56] utilize CNN and GAN to extract common regions from unregistered images before performing CD, addressing the challenges posed by unregistered bitemporal images. DFGAN [57] utilizes GAN based on federated learning to achieve privacy preservation. GSIA [58] proposes a GAN-based data augmentation network, addressing the issue of insufficient correlation between the training and test sets in domain adaptation. ChangeAnywhere [59] utilizes a semantically latent diffusion model and single-temporal images to generate CD samples, forming the currently largest synthetic CD dataset. PSA [60] integrates SAM and CLIP with CD, enabling the introduction of semantic understanding via simple texts to filter out pseudochanges, achieving unsupervised learning for CD.

Unlike the problems addressed in these methods, we focus on solving how to avoid feature confusion while maintaining computational efficiency when tackling spatial–temporal inconsistency. Accordingly, we propose a novel CD framework based on STRobustNets.

III. Method

We propose a novel CD framework based on STRobustNets. The overall procedure of our STRobustNet is illustrated in Fig. 2. Our method starts by extracting multiscale feature maps (multiscale feature extraction module). Then, the feature maps are sent to extract STRobustNets for each land-use class of interest (RRGModule). Next, these representations are used to generate spatial–temporal consistent classification representations for CD [change map prediction module (CMPModule)]. Finally, we supplement the details of the low-resolution change map according to the original-resolution bitemporal images, facilitating more precise predictions [detail refinement module (DRModule)]. We refer to the model without the DRModule as

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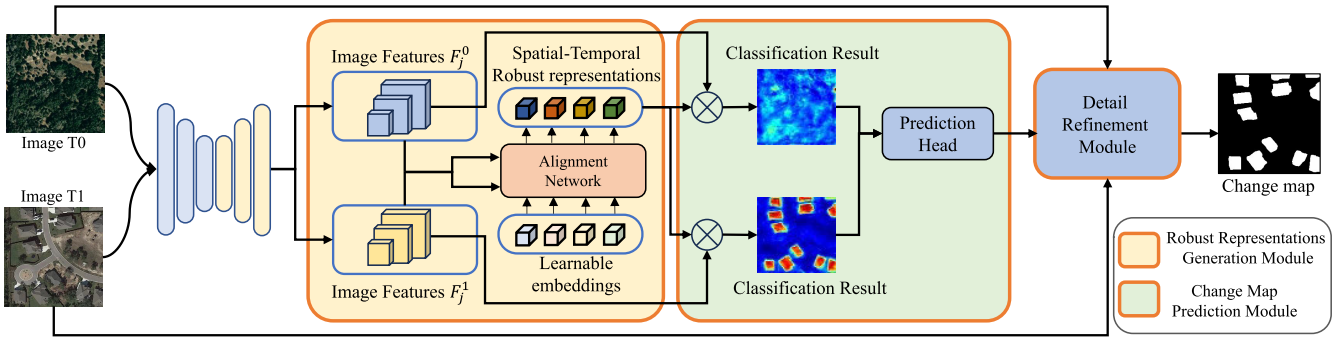


Fig. 2. Architecture of the proposed STRobustNet. ⊗ means the classification operation.

the fast version of our model, which we named STRobustNet (fast).

We will detail the multiscale feature extraction module and the loss function in Sections III-A and III-E, respectively. The proposed RRGModule and CMPModule are introduced in Sections III-B and III-C, respectively. The designed DRModule is described in Section III-D.

A. Multiscale Feature Extraction Module

In remote sensing images, objects exhibit different sizes, making multiscale features crucial. Therefore, we utilize the multiscale feature encoder module (MSFEModule) to extract multiscale feature maps. First, we employ ResNet-18 [61] to extract features corresponding to 1/4, 1/8, and 1/16 of the original image size, denoted as F_2^t, F_1^t, and F_0^t, where $t = 0, 1$ refers to the image at time t.

Second, we combine the features from adjacent scales

$$\boldsymbol{y}_j^t = \begin{cases} \hat{\boldsymbol{y}}_j^t, & j = 0 \\ \hat{\phi}_j\left(\hat{\boldsymbol{y}}_j^t\right), & j = 1, 2 \end{cases} \tag{1}$$

$$\hat{\boldsymbol{y}}_j^t = \begin{cases} \phi_j\left(F_j^t\right), & j = 0 \\ \phi_j\left(F_j^t\right) + \boldsymbol{y}_{j-1}^t \uparrow, & j = 1, 2 \end{cases} \tag{2}$$

where $\boldsymbol{y}_j^t \in \mathbb{R}^{C\times(H/2^{(4-j)})\times(W/2^{(4-j)})}$ represents the fused multiscale feature maps and C represents the number of feature channels. The operator $\hat{\phi}_j$ represents a single-layer convolution operation with a 3×3 convolutional kernel and a stride of 1, followed by a batch normalization operation. ϕ_j denotes a single-layer convolution operation with a 1×1 convolutional kernel and a stride of 1. The symbol $\uparrow$ denotes the bilinear interpolation upsampling operation that increases the spatial resolution by a factor of 2. The fused multiscale features are subsequently utilized in the alignment network to refine the STRobustNets.

B. Robust Representation Generation Module

The goal of RRGModule is to calculate a set of STRobustNets for land-use classes of interest, which unify the feature representations of the same land-use class across different times and regions. Therefore, RRGModule considers both the universal spatial–temporal context from the entire dataset and the specific spatial–temporal context from the current bitemporal images.

As illustrated in Fig. 2, we first initialize a set of learnable embeddings $\boldsymbol{e} \in \mathbb{R}^{N_r\times C}$ to learn the universal robust representations of the interested land-use classes, where N_r is the number of the representations. These universal robust representations (i.e., the updated $\boldsymbol{e}$) play an indirect but crucial role in generating the change map, making them intricately linked to the overall loss function $\mathcal{L}$. Consequently, we can compute the gradient of the loss function with respect to these embeddings as $(\partial_{\mathcal{L}}/\partial_{\boldsymbol{e}})$. In other words, the universal robust representations can be obtained in a data-driven manner, and we do not need any extra supervision.

Due to variations in imaging conditions across different bitemporal images, the classification representations obtained by universal robust representations on specific images may not achieve optimal results. Therefore, we align the universal representations to the current bitemporal images using the alignment network shown in Fig. 3, which mitigates the impact of varying imaging conditions. The alignment network includes L alignment layers (ALs), each comprising three alignment blocks. The details of the alignment block are given as follows.

Step 1: Given $\boldsymbol{y}_j^0$ and $\boldsymbol{y}_j^1$ of the multiscale bitemporal features from the current bitemporal images $\mathbf{I}_0$ and $\mathbf{I}_1$, we first reshape the features into a sequence $\boldsymbol{p}_j^t = [p_{j_1}^t, \ldots, p_{j_N}^t]$, where N represents the number of the pixels in $\boldsymbol{y}_j^0$ or $\boldsymbol{y}_j^1$.

To better utilize the temporal context of the current scene, we concatenate $\boldsymbol{p}_j^0$ and $\boldsymbol{p}_j^1$ along the spatial dimension

$$\boldsymbol{p}_j = \text{concat}\left(\boldsymbol{p}_j^0, \boldsymbol{p}_j^1\right) \tag{3}$$

where $p_j \in \mathbb{R}^{2N\times C}$.

Step 2: We use the universal representation as the query to retrieve relevant features from the scene described by $\boldsymbol{p}_j$ based on the multihead cross-attention (MCA) operator [62] and fuse them into the universal representations

$$\begin{aligned} \hat{\boldsymbol{z}}_{j+1}^l &= \mathbf{MCA}\left(\boldsymbol{z}_j^l + \boldsymbol{e}, \boldsymbol{p}_j + E, \boldsymbol{p}_j\right) + \boldsymbol{z}_j^l \\ \boldsymbol{z}_{j+1}^l &= \mathbf{LN}\left(\mathbf{MLP}\left(\mathbf{LN}\left(\hat{\boldsymbol{z}}_{j+1}^l\right)\right) + \hat{\boldsymbol{z}}_{j+1}^l\right) \quad j \in \{0, 1, 2\} \end{aligned} \tag{4}$$

where $E \in \mathbb{R}^{N\times C}$ represents the Fourier position embedding, and $\boldsymbol{z}_{j+1}^l$ represents the output of the jth block at layer l, while $\boldsymbol{z}_j^l$ is the input of the jth block. $\mathbf{LN}(\cdot)$ denotes the layer normalization. $\mathbf{MLP}(\cdot)$ represents the multilayer perceptron comprising two linear transformation layers with an ReLU activation [63] in between. Note that we compute the sum

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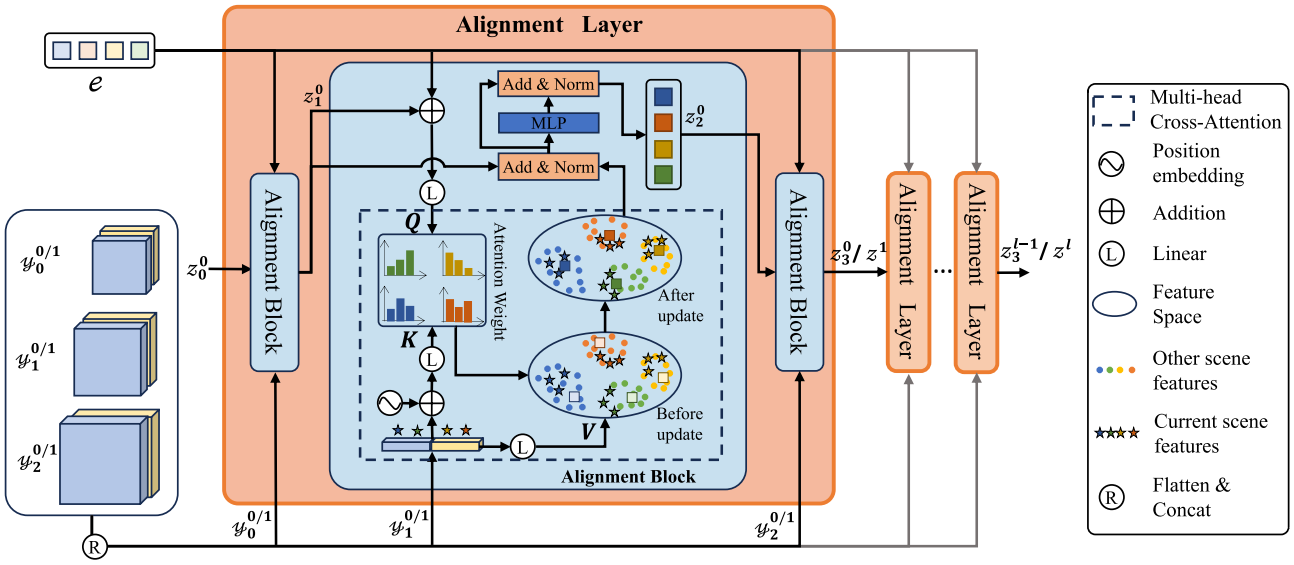


Fig. 3. Illustration of the alignment network in the proposed RRGModule. "Current scene features" refer to the features extracted from the input image pairs that require CD, while "other scene features" denote the features of the remaining image pairs in the dataset, excluding the current input.

of the output z_j^l from the previous alignment block and the STRR-Us $\boldsymbol{e}$ as the input to each alignment block. This step is vital for preserving the universal statistical properties of the representation during the alignment process.

MCA in (4) takes a triplet of query, key, and value as input. Based on the attention between the query and key, features are selected from the value and integrated into the query. Formally, **MCA** is defined as follows:

$$\begin{aligned}\mathbf{MCA}(\mathbf{Q},\mathbf{K},\mathbf{V}) &= \text{concat}(\text{head}_1,\ldots,\text{head}_H)\mathbf{W}^O \\ \text{head}_h &= \mathbf{Att}\left(\mathbf{Q}\mathbf{W}_h^q, \mathbf{K}\mathbf{W}_h^k, \mathbf{V}\mathbf{W}_h^v\right)\end{aligned} \tag{5}$$

where $\mathbf{W}_h^q, \mathbf{W}_h^k \in \mathbb{R}^{C\times C_k}$ and $\mathbf{W}_h^v \in \mathbb{R}^{C\times C_v}$ are the learnable linear projection weights at head h, which can project the queries, keys, and values to C_k, C_k, and C_v dimensions. $\mathbf{W}_O \in \mathbb{R}^{C\times HC_v}$ is also a group of linear projection weights, in which H represents the number of attention heads. In this article, we use $C_k = C_v = C/H$. **Att** denotes the attention mechanism. Formally,

$$\mathbf{Att}(\mathbf{Q},\mathbf{K},\mathbf{V}) = \sigma\left(\frac{\mathbf{Q}\mathbf{K}^T}{\sqrt{d}}\right)\mathbf{V} \tag{6}$$

where $\sigma(\cdot)$ denotes the softmax function, d is the scale factor, and T is the matrix transpose operation.

In each alignment layer l, we iterate j from 0 to 2. That is, take z_0^{l-1} as the input and sequentially pass it through 3 structure-identical alignment blocks, resulting in the output z_3^{l-1}/z^l. The original input z_0^0 is a zero array. The universal representations pass through L ALs sequentially, obtaining aligned specific representations z.

C. Change Map Prediction Module

CMPModule calculates the spatial–temporally consistent classification representation and detects changes on these classification representations.

First, we activate the bitemporal feature $\boldsymbol{y}_0^t$ with each robust representation in $\boldsymbol{z}$ as follows:

$$A_t = \sigma\left(\boldsymbol{z}\boldsymbol{y}_0^{t\,T}\right) \tag{7}$$

where $A_t \in \mathbb{R}^{N_r\times H/4\times W/4}$ denotes the multichannel activation maps at time t, with the number of channels corresponding to the number of representations. σ represents the softmax function, which is used to normalize the class dimension. T signifies matrix transpose operations. Each activation value $A_t(x, y, n)$ quantifies the extent to which the corresponding pixel $\boldsymbol{y}_0^t(x, y)$ belongs to the land-use representation $\boldsymbol{z}(n)$ activating this pixel. Here, n indexes the robust representations of different land-use classes, and (x, y) is the pixel position. In this way, each pixel is represented by a set of activation values indicating its land-use class, i.e., classification representation.

Then, A_0 and A_1 are upsampled to the original resolution to obtain $A_0^*, A_1^* \in \mathbb{R}^{N_r\times H\times W}$. A_0^*, A_1^* is concatenated along the class dimension and input into the prediction head. The prediction head is used to generate the logits map $\hat{\mathbf{P}} \in \mathbb{R}^{2\times H\times W}$

$$\hat{\mathbf{P}} = f\left(\text{concat}\left(A_0^*, A_1^*\right)\right) \tag{8}$$

where f: $\mathbb{R}^{2N_r\times H\times W} \to \mathbb{R}^{2\times H\times W}$ is the prediction head that consists of two 1×1 convolutional layers with BatchNorm and ReLU in between. The output channel number of the last convolution is 2. $\hat{\mathbf{P}}$ will be input to the DRModule for detail enhancement. Note that in STRobustNet (fast), the change mask $\mathbf{M} \in \mathbb{R}^{H\times W}$ is computed by a pixelwise argmax operation on the channel dimension of $\hat{\mathbf{P}}$.

D. Detail Refinement Module

Due to the utilization of low-resolution feature maps for aligning the representations and performing classification, the resulting change map may contain slight errors in the delicate regions. Inspired by previous work [64], [65], we propose

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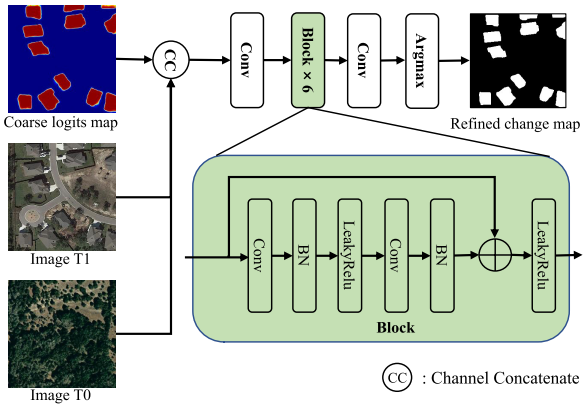


Fig. 4. Illustration of our proposed DRModule. ⊕ means elementwise summation. The "coarse logits map" is a two-channel probability map output by STRobustNet (fast), where one channel indicates the probability of a change region and the other indicates the probability of no change.

the DRModule that utilizes fine-grained guidance from the original bitemporal images to repair the details of the change regions.

The structure of our DRModule is illustrated in Fig. 4. After obtaining the logits map $\hat{\mathbf{P}}$, we concatenate the original high-resolution bitemporal images with $\hat{\mathbf{P}}$ along the channel dimension. Then, they are input into a convolution layer, expanding the channel dimension from 8 to 32. Next, the feature maps of 32 channels are input into six convolutional blocks for information fusion. Each block consists of two sets of "conv → BN → LeakyReLU" structures, with a residual connection between the input of the block and the output of the last BN. The output channels of each convolution layer in the block remain 32. Finally, a convolutional layer will be used on the feature maps output from the last block to generate the two-channel logits map $\hat{\mathbf{P}}^* \in \mathbb{R}^{2\times H\times W}$. We perform a pixelwise argmax operation on the channel dimension of $\hat{\mathbf{P}}^*$ to calculate the refined change map.

E. Loss Function

The loss function comprises the intersection over union (IoU) loss $\mathcal{L}_{\text{IoU}}$, the edge-enhanced cross-entropy loss $\mathcal{L}_{\text{edge}}$, and the smoothness loss $\mathcal{L}_{\text{smooth}}$. Before computing the loss, we apply softmax along the channel dimension to the logits $\hat{\mathbf{P}}^*$ (or $\hat{\mathbf{P}}$ for the fast version) to generate the change probability maps $\mathbf{P}^*$ (or $\mathbf{P}$).

Similar to most existing methods, we employ IoU loss and cross-entropy loss to constrain the logits. The IoU loss is

$$\mathcal{L}_{\text{IoU}} = 1 - \frac{\sum_p \mathbf{x}_p \mathbf{t}_p}{\sum_p \left(\mathbf{x}_p + \mathbf{t}_p - \mathbf{x}_p \mathbf{t}_p\right)} \tag{9}$$

where p denotes pixel, $\mathbf{t}_p$ denotes the label of p, and $\mathbf{x}_p \in (\mathbf{P}$ or $\mathbf{P}^*)$, which denotes the probability that pixel p is change.

The area within an object is significantly larger than that of its boundary. Consequently, directly employing the existing cross-entropy loss often causes the network to prioritize optimizing the object's interior, leading to increased ambiguity in the predicted $\mathbf{P}^*$ at the boundaries. To balance this disproportion and better guide the DRModule to extract effective information from bitemporal images for refining edges, we introduce an edge-enhanced cross-entropy loss

$$\mathcal{L}_{\text{edge}} = -\frac{1}{N}\sum_p W(p)\left(\mathbf{t}_p \log\left(\mathbf{x}_p\right) + \left(1-\mathbf{t}_p\right)\log\left(1-\mathbf{x}_p\right)\right) \tag{10}$$

where N denotes the number of p. $W(p)$ is the loss weight assigned to p, which involves applying the Laplace operator to filter the ground truth, followed by a mapping of the resultant values to the interval [0.5, 1) through the sigmoid function. W in the edge regions will be larger than that in the other regions.

While guiding the network to focus on boundaries, it may introduce prediction noise within the object's interior. In addition, to constrain the DRModule and prevent it from introducing redundant information from bitemporal images, we simultaneously introduce a smoothness loss [66], [67]

$$\mathcal{L}_{\text{smooth}} = \frac{1}{N}\sum_p \left|\partial_h \mathbf{x}_p\right| e^{-\left\|\partial_h \mathbf{t}_p\right\|_1} + \left|\partial_w \mathbf{x}_p\right| e^{-\left\|\partial_w \mathbf{t}_p\right\|_1} \tag{11}$$

where ∂_h and ∂_w represent the gradients of $\mathbf{P}^*$ and the ground truth calculated in the h and w directions, respectively.

The total loss can be formulated as follows:

$$\mathcal{L} = \mathcal{L}_{\text{edge}} + \mathcal{L}_{\text{smooth}} + \alpha \mathcal{L}_{\text{IoU}} \tag{12}$$

where α controls the importance of L_{IoU} as a hyperparameter.

IV. Experiments

A. Datasets

LEVIR-CD [13]: It contains 647 pairs of ultrahigh-resolution (VHR 0.5 m/pixel) Google Earth images with a size of 1024 × 1024. These bitemporal images cover a time span of 5–14 years. The dataset focuses on building-related changes and covers various building types such as villa homes, high-rise apartments, small garages, and large warehouses. We use the authors' default dataset division and crop images into patches of size 256 × 256 without overlap, obtaining 7120/1024/2048 pairs of patches for training/validation/testing, respectively.

WHU-CD [14]: It contains a pair of high-resolution (0.075 m) aerial images with the size of 32 507 × 15 354. The dataset focuses on building-related changes. We divide the images into nonoverlapping patches of size 256 × 256 and randomly divide them into 6096/762/762 training/validation/testing pairs.

SYSU-CD [68]: It contains 20 000 pairs of aerial images with a size of 256 × 256 and resolution of 0.5 m, which were taken in Hong Kong from 2007 to 2014. The main types of changes include: 1) newly built urban buildings; 2) suburban dilation; 3) groundwork before construction; 4) change of vegetation; 5) road expansion; and 6) sea construction. We use the original split of the dataset, which has 12 000/4000/4000 pairs of patches for training/validation/testing, respectively.

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B. Evaluation Metrics

We use TP, TN, FP, and FN to represent the number of true positives, true negatives, false positives, and false negatives, respectively. The evaluation metrics, **F1-score (F1)**, **IoU**, **overall accuracy (OA)**, **precision (Pre)**, and **recall (Rec)**, used in this article can be calculated according to the following equations:

$$\mathbf{F1} = 2 \times (\mathbf{Pre} \times \mathbf{Rec})/(\mathbf{Pre} + \mathbf{Rec}) \tag{13}$$

$$\mathbf{IoU} = \mathrm{TP}/(\mathrm{TP} + \mathrm{FN} + \mathrm{FP}) \tag{14}$$

$$\mathbf{OA} = (\mathrm{TP} + \mathrm{TN})/(\mathrm{TP} + \mathrm{TN} + \mathrm{FN} + \mathrm{FP}) \tag{15}$$

$$\mathbf{Pre} = \mathrm{TP}/(\mathrm{TP} + \mathrm{FP}) \tag{16}$$

$$\mathbf{Rec} = \mathrm{TP}/(\mathrm{TP} + \mathrm{FN}). \tag{17}$$

During the evaluation process, we compute the confusion matrix for each predicted result against the ground truth and then sum up all the confusion matrices within each testing set. Finally, we calculate the values of each evaluation metric based on the summed confusion matrix.

C. Implementation Details

Our models are implemented on PyTorch and trained using a single NVIDIA A40 GPU. We apply data augmentation to the input image patches, including random flip, random rescale, random crop, Gaussian blur, and random color jitter. We use stochastic gradient descent (SGD) with a momentum of 0.9 and a weight decay of 0.0005 to optimize the model. In the LEVIR-CD and WHU-CD datasets, we set the learning rate to 0.01, configure four learnable embeddings and one alignment layer, and set the IoU loss weight α to 0.5. In the SYSU-CD dataset, we set the learning rate to 0.005 and configure 20 learnable embeddings and one alignment layer. The learning rate will decay linearly to 0 within 200 epochs of training. Due to the coarse ground-truth annotations at change boundaries in the SYSU-CD dataset, overly strong boundary constraints or point-to-point constraints may confuse the model. Therefore, we only use normal cross-entropy loss, smoothness loss, and IoU loss to train the model. Moreover, we assign a higher weight to the IoU loss, i.e., $\alpha = 0.7$, to strengthen the constraint on the overlap between the predictions and the ground-truth annotations for the SYSU-CD dataset.

D. Performance Comparison on Benchmarks

In this section, we analyze the advantages of our proposed STRobustNet compared to existing top-performing methods from both efficiency and accuracy perspectives.

1) Analysis of Complexity: We compare the computational complexity of different methods based on the number of parameters (Params.), FLOPs, and inference speed (FPS) in Table I. Because of the distinctive encapsulation in MaskCD's code, we could not calculate its FLOPs with the "fvscore" tool. In addition, we cannot access the FPS of MFINet as it is not open source.

First, compared to CNN-based methods, our methods have a higher computational complexity, but both STRobustNet (fast) and STRobustNet provide significant advantages in terms of detection accuracy. Then, compared to transformer-based methods, STRobustNet (fast) has the lowest FLOPs and the highest FPS. Despite having more parameters compared to HANet and MFINet, our method exhibits fewer FLOPs and faster inference speed. In addition, based on STRobustNet (fast), we introduce a lightweight DRModule and obtain the final STRobustNet. This additional module significantly improves the accuracy while only increasing the parameter count by 2%. At the same time, STRobustNet maintains its advantage in terms of inference speed, with a speed of 31.11 FPS, compared to other top-performing transformer-based methods. Fig. 5 provides a more intuitive visualization of the relationship between **F1** and the number of parameters, and **F1** and FPS. Based on the figure, it is evident that *STRobustNet exhibits significant advantages in terms of computational efficiency and maintains competitive accuracy.*

2) Analysis of Accuracy: We comprehensively analyze the accuracy of STRobustNet quantitatively and qualitatively. Experimental results demonstrate that STRobustNet achieves excellent overall performance comparable to existing methods.

First, we compare STRobustNet with CNN-based methods. CNN-based methods could not effectively handle the issue of spatial–temporal inconsistency as mentioned in the related work, resulting in a higher rate of false detections. For

TABLE I

COMPARISON OF MODELS' COMPLEXITY. THE TOP THREE RESULTS ARE HIGHLIGHTED IN **RED**, **GREEN**, AND **BLUE**. THE SYMBOL "-" INDICATES THAT WE ARE UNABLE TO ACCESS THE CORRESPONDING DATA. METHOD NAMES SUFFIXED WITH "*" INDICATE THAT THEY ARE CNN-BASED METHODS; OTHERWISE, THEY ARE TRANSFORMER-BASED METHODS. THE SIZE OF THE INPUT IMAGE TO THE MODEL IS 256 × 256 × 3 TO CALCULATE THE FLOPS AND FPS

model	Year	F1↑	Params.(M)↓	FLOPs(G)↓	FPS↑
FC-EF*[29]	2018	83.40	1.35	3.57	76.22
FC-Siam-Di*[29]	2018	86.31	1.35	4.72	56.72
FC-Siam-Conc*[29]	2018	83.69	1.55	5.33	56.66
STANet[13]	2020	87.26	16.93	18.03	23.86
BIT[19]	2022	89.31	12.40	10.87	30.61
Changeformer[43]	2022	90.40	41.03	45.97	13.04
EGCTNet[44]	2022	90.68	106.16	33.52	6.23
HANet[20]	2023	90.28	3.02	14.07	22.30
VcT[2]	2023	90.04	12.41	10.64	21.15
MFINet[69]	2024	90.12	4.95	6.89	-
EATDer[70]	2024	89.75	6.61	23.46	21.15
BiFA[17]	2024	**90.69**	5.58	53.00	17.89
MaskCD[47]	2024	**90.94**	107.40	-	12.05
STRobustNet (fast)	2024	89.92	5.11	5.71	37.45
STRobustNet	2024	**91.11**	5.23	13.19	31.11

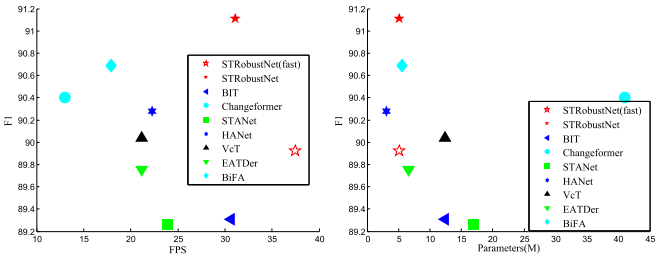


Fig. 5. Comparison with SOTA methods on the LEVIR-CD dataset. Left: FPS versus F1-score. Right: model parameters versus F1-Score.

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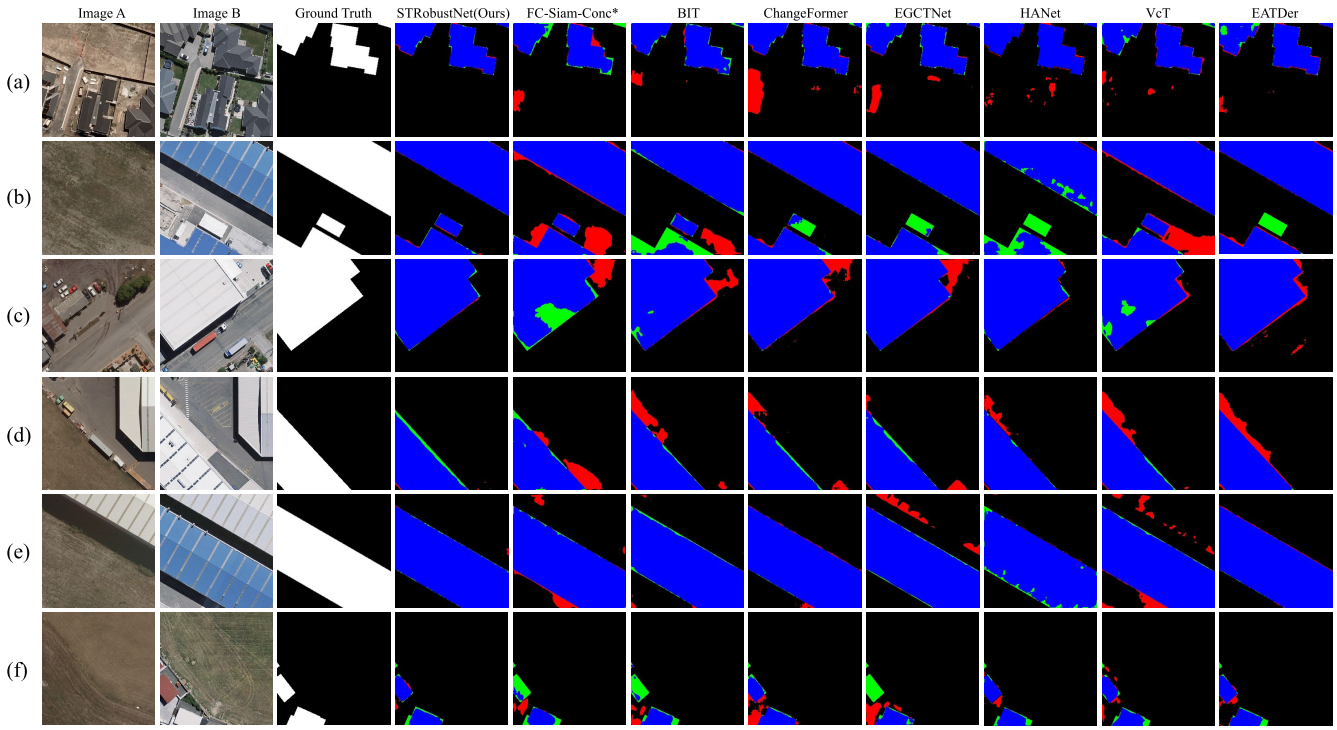


Fig. 6. Visualization of some state-of-the-art methods' performance on the WHU-CD dataset. The color scheme employed for visualization is given as follows: **blue** represents **true positives**, **black** represents **true negatives**, **red** represents **false positives**, and **green** represents **false negatives**. Method names suffixed with "*" indicate that they are CNN-based methods; otherwise, they are transformer-based methods.

example, the dataset shown in Fig. 6 contains instances where the same land-use class undergoes significant changes due to temporal differences. In such cases, CNN-based methods tend to misclassify regions of the same land-use class with significant differences as changed areas, as shown by the red areas (false positive regions) in Fig. 6. In contrast, our method overcomes this problem by detecting changes based on spatial–temporal consistent classification representations obtained from the STRobustNets. As a result, our predictions have fewer red areas in Fig. 6 and exhibit significant advantages in terms of accuracy compared with CNN-based methods, as shown in Tables II and III. Note that our method has a lower **precision** compared to the "FC-Siam-Conc" in Table II. However, this does not necessarily indicate that our method is inferior. **Precision** solely represents the accuracy of detected changed regions. In addition, "FC-Siam-Conc" exhibits significantly lower **recall** and IoU compared with our method. Thus, the low **recall** of the "FC-Siam-Conc" method leads to its higher **precision**. This can be observed in Fig. 7, where the predictions of "FC-Siam-Conc" show a significant number of false negative regions (green areas) compared to our method.

Second, we compare STRobustNet with transformer-based methods. Fig. 7 presents the predictions of various methods on the LEVIR-CD dataset. This dataset contains numerous instances where different classes have similar features, such as buildings and soil in samples [see Fig. 7(c) and 7(d)]. Due to feature confusion, transformer-based methods often misclassify these similar but different land-use classes as unchanged regions. As a consequence, a significant area of FN (green areas) can be observed in the predictions of transformer-based methods shown in Fig. 7. In contrast, our STRobustNet can mitigate the impact of feature confusion, resulting in superior overall performance compared to transformer-based methods in such scenarios. On the one hand, from a qualitative analysis standpoint, as shown in both Figs. 6 and 7, the areas of FN regions (green areas) and FP regions (red areas) in our

TABLE II

COMPARISON RESULTS ON THE LEVIR-CD DATASET. THE TOP THREE RESULTS ARE HIGHLIGHTED IN **RED**, **GREEN**, AND **BLUE**. ALL THE SCORES ARE DESCRIBED IN PERCENTAGE (%). THE SYMBOL "-" INDICATES THAT THE METHOD IS NOT OPEN SOURCE, AND WE ARE UNABLE TO ACCESS THE CORRESPONDING DATA. METHOD NAMES SUFFIXED WITH "*" INDICATE THAT THEY ARE CNN-BASED METHODS; OTHERWISE, THEY ARE TRANSFORMER-BASED METHODS

model	Year	OA↑	F1↑	IoU↑	Pre↑	Rec↑
FC-EF*[29]	2018	98.39	83.40	71.53	86.91	80.17
FC-Siam-Di*[29]	2018	98.67	86.31	75.92	89.53	83.31
FC-Siam-Conc*[29]	2018	98.49	83.69	71.96	91.99	76.77
DTCDSCN*[31]	2020	98.77	87.67	78.05	88.53	86.83
STANet[13]	2020	98.66	87.26	77.40	83.81	**91.00**
BIT[19]	2022	98.92	89.31	80.68	89.24	89.37
Changeformer[43]	2022	99.04	90.40	82.48	92.05	88.80
Hybrid-TransCD[1]	2022	99.00	90.06	81.92	91.45	88.72
MSTDSNet[46]	2022	-	88.10	78.73	85.52	**90.84**
EGCTNet[44]	2022	99.06	90.68	82.95	**92.20**	89.21
HANet[20]	2023	99.02	90.28	82.27	91.21	89.36
VcT[2]	2023	99.01	90.04	81.89	**92.57**	87.65
DMMSTNet[71]	2023	99.08	90.83	83.20	92.11	89.59
MFINet[69]	2024	**99.21**	90.12	82.33	92.09	89.02
EATDer[70]	2024	98.92	89.75	81.40	86.93	**92.75**
BiFA[17]	2024	99.06	**90.69**	**82.96**	91.52	89.86
MaskCD[47]	2024	**99.09**	**90.94**	**83.38**	**92.50**	89.43
STRobustNet (fast)	2024	98.99	89.92	81.69	91.61	88.29
STRobustNet	2024	**99.10**	**91.11**	**83.66**	91.54	90.67

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TABLE III

COMPARISON RESULTS ON WHU-CD AND SYSU-CD DATASETS. THE TOP THREE RESULTS ARE HIGHLIGHTED IN **RED**, **GREEN**, AND **BLUE**. ALL THE SCORES ARE DESCRIBED IN PERCENTAGE (%). METHOD NAMES SUFFIXED WITH "*" INDICATE THAT THEY ARE CNN-BASED METHODS; OTHERWISE, THEY ARE TRANSFORMER-BASED METHODS

model	Year	WHU-CD					SYSU-CD					FLOPs(G)	FPS
		OA	F1	IoU	Pre	Rec	OA	F1	IoU	Pre	Rec		
FC-EF*[29]	2018	98.35	76.31	61.70	84.40	69.64	86.02	75.07	60.09	74.32	75.84	3.57	76.22
FC-Siam-Di*[29]	2018	98.70	82.13	69.68	86.55	78.15	82.11	72.57	59.96	**89.13**	61.21	4.72	56.72
FC-Siam-Conc*[29]	2018	98.87	84.65	73.39	88.06	81.50	86.17	76.35	61.75	82.54	71.03	5.33	56.66
STANet[13]	2020	-	-	-	-	-	87.96	77.37	63.09	70.76	**85.33**	18.03	23.86
BIT[19]	2022	98.87	85.46	74.62	83.91	87.07	90.18	78.15	64.13	74.49	**82.18**	10.87	30.61
Changeformer[43]	2022	99.28	90.45	82.56	91.34	89.58	90.44	78.45	64.54	83.73	73.80	45.97	13.04
EGCTNet[44]	2022	99.27	90.13	82.03	92.91	87.50	90.90	80.37	67.18	81.82	78.97	33.52	6.23
HANet[20]	2023	99.22	90.13	82.03	92.26	88.09	89.52	77.41	63.14	78.71	76.14	14.07	22.30
VcT[2]	2023	99.04	87.63	77.98	86.29	89.01	90.08	79.08	65.40	78.67	79.50	10.64	21.15
EATDer[70]	2024	99.24	90.58	82.79	87.82	**93.52**	90.34	80.23	66.99	77.56	**83.09**	23.46	21.15
BiFA[17]	2024	**99.48**	**93.11**	**87.11**	**94.35**	**91.90**	**92.75**	**84.12**	**72.59**	**86.98**	81.44	53.00	17.89
MaskCD[47]	2024	**99.57**	**94.42**	**89.43**	**94.31**	**94.53**	**92.34**	**82.93**	**70.84**	**87.41**	78.89	-	12.05
STRobustNet	2024	**99.32**	**90.89**	**83.29**	**92.92**	88.94	**91.13**	**80.66**	**67.59**	82.98	78.46	13.19	31.11

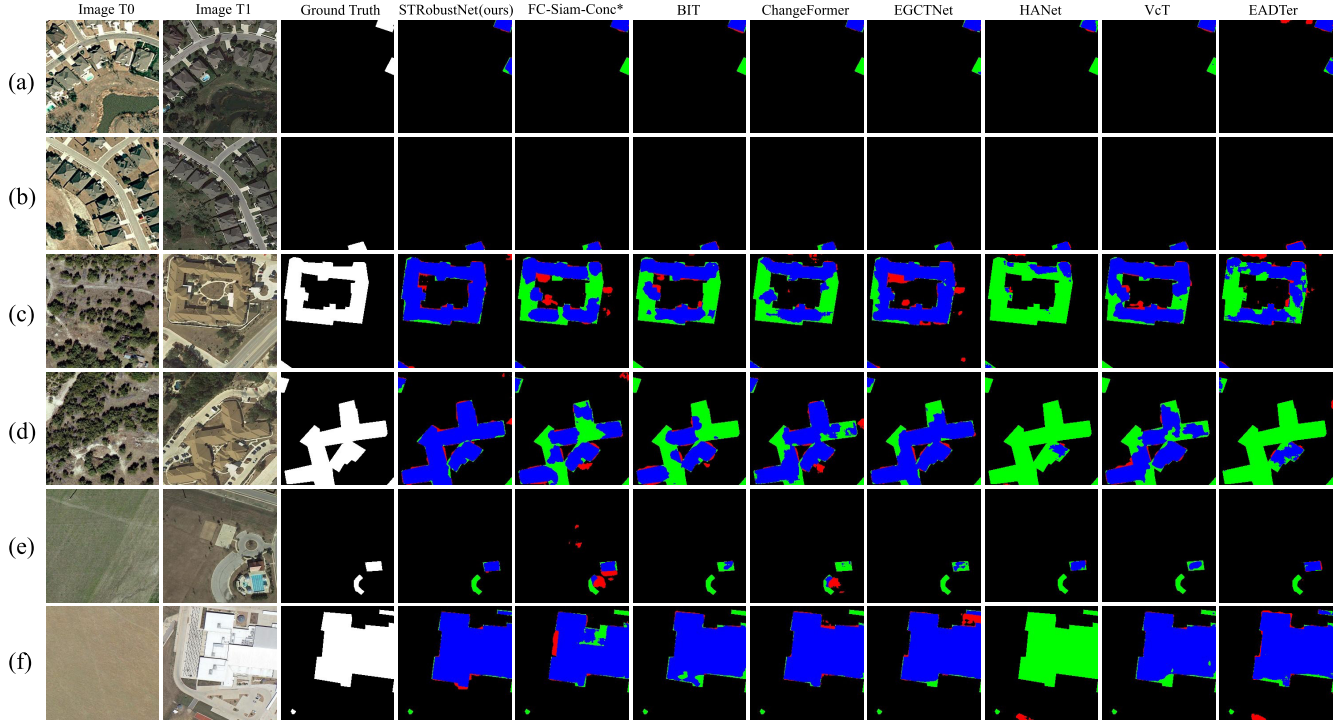


Fig. 7. Visualization of some state-of-the-art methods' performance on the LEVIR-CD dataset. The color scheme employed for visualization is given as follows: **blue** represents **true positives**, **black** represents **true negatives**, **red** represents **false positives**, and **green** represents **false negatives**. Method names suffixed with "*" indicate that they are CNN-based methods; otherwise, they are transformer-based methods.

method's predictions are minimal compared to the transformer-based methods. On the other hand, from a quantitative analysis standpoint, STRobustNet demonstrates superior overall performance compared to transformer-based methods. Although STRobustNet has slightly lower **recall** or **precision** according to the results in Tables II and III, it achieves a better balance between these two metrics compared with transformer-based methods. For instance, in Table III, compared to the method with the highest **recall** (STANet), our |log(precision/recall)| value is 0.01, while STANet's |log(precision/recall)| value is 0.81. Most importantly, STRobustNet achieves the highest **F1** and IoU on the LEVIR-CD dataset.

Note that due to our pursuit of a balance between efficiency and accuracy, in Table III, our method is slightly inferior to BiFA [17] and MaskCD [47] of high computational complexity. Compared to our proposed STRobustNet, BiFA and MaskCD employ a more complex backbone for feature extraction, whereas we use a lightweight ResNet-like network, resulting in relatively weaker feature representation. In addition, MaskCD utilizes the pretrained model of Mask2Former on ADE20K for network training. The powerful segmentation capabilities of the pretrained Mask2Former enhance MaskCD's performance in segmenting changed regions based on change-aware features. BiFA introduces a differential flow field to mitigate inadequate registration of bitemporal images before performing CD, which is overlooked in our method. Therefore, BiFA and MaskCD outperform our method on the WHU-CD and SYSU-CD datasets. However, as shown in Table I, the FLOPs of BiFA are more than four times those of STRobustNet, and the parameters of MaskCD exceed

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TABLE IV

ABLATION STUDY ON THE LEVIR-CD DATASET. STRR-U: UTILIZING STROBUSTNETS DERIVED FROM THE ENTIRE DATASET. STRR-S: UTILIZING STROBUSTNETS DERIVED FROM THE SPECIFIC BITEMPORAL IMAGES. ALIGN: ALIGNING STROBUSTNETS WITH CURRENT BITEMPORAL FEATURES. MS: UTILIZING MULTISCALE BITEMPORAL FEATURES FOR ALIGNMENT. DR: EMPLOYING THE DRMODULE. THE HIGHEST SCORE IS MARKED IN BOLD

Base	STRR-U	STRR-S	Align	MS	DR	OA	F1	IoU
✓						98.74	87.90	78.42
✓	✓					98.97	89.55	81.08
✓		✓				98.91	88.98	80.15
✓	✓		✓			98.97	89.71	81.34
✓	✓		✓	✓		98.99	89.92	81.69
✓	✓		✓	✓	✓	**99.10**	**91.11**	**83.66**

that of STRobustNet by over 20 times, while both of them significantly lag behind STRobustNet in terms of speed. Consequently, our method is more lightweight and has greater potential for real-time CD on edge devices.

E. Ablation Study

In this section, we design a baseline model, referred to as Base, to investigate the roles of different components in STRobustNet. The Base uses the same feature extractor as STRobustNet but only utilizes bitemporal features at the maximum scale. The subtracted bitemporal features are directly fed into the prediction head for CD. Due to the lack of handling temporal inconsistency in Base, its accuracy is relatively lower, as shown in Table IV.

1) Impact of Different Components in the RRGModule:

1) *Universal Representations:* We introduce the learnable embeddings into Base to capture the STRR-U of different land-use classes, which are then used to activate the bitemporal features for CD. As shown in the first and second rows of Table IV, utilizing STRR-Us leads to an improvement in accuracy. Because learnable embeddings have learned spatial–temporal consistent representations, providing classification representations with stronger consistency compared to the original bitemporal features. Furthermore, we compute the representations (STRR-S) solely from the currently input bitemporal images following the BIT method [19], replacing the previously used universal representations. The results, shown in the third row, indicate a decline in performance across all metrics compared with the second row. This demonstrates that the universal spatial–temporal contexts provided by the entire dataset contribute more to the robustness of the representation than the specific contexts derived from a single pair of bitemporal images.
2) *Alignment Network:* We align the universal representations to the current bitemporal features by the designed alignment network. As illustrated in the third row of Table IV, the alignment operation leads to an improvement in accuracy. This demonstrates the alignment operation is effective in mitigating the deviation between the universal representations and the specific input scene.

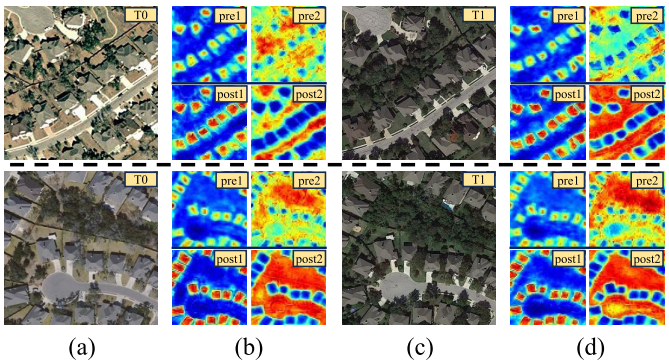


Fig. 8. Visualization of alignment operation's effectiveness. (a) Bitemporal image T0. (b) Regions of interest indicated by the robust representations before and after alignment on image T0. (c) Bitemporal image T1. (d) Regions of interest indicated by the robust representations before and after alignment on image T1. Pre: before alignment; post: after alignment. (b) and (d) Derived through calculating the dot product between each robust representation of STRobustNet and the pixel-level features extracted from bitemporal images.

Furthermore, to adapt the robust representations to diverse object sizes, we introduce multiscale features to the alignment layer. The results in the fourth row of Table IV show improvement compared to the third row, validating the effectiveness of multiscale alignment.

To better understand the effect of the alignment operation, we visualize the activation maps before and after alignment in Fig. 8. Even though the STRobustNets before alignment can activate the correct regions, the activation values are not sharp due to slight differences from the current specific sample. However, after alignment, the activation values of the representations clearly differentiate between buildings and the surrounding environment.

Furthermore, we visualize the distribution of the four robust representations in LEVIR-CD before and after the alignment network. Since we lack category labels, we manually select 58 pairs of bitemporal images containing buildings and divide them into building areas (purple) and nonbuilding areas (green) based on change labels. Next, we choose one pair of images from the 58 pairs and align the universal robust representations to this pair of images by the alignment network, obtaining the specific robust representations. We feed the features of building and nonbuilding areas from all 58 pairs of images, along with universal robust representations and specific robust representations, into t-SNE for visualization. As shown in Fig. 9, one robust representation (red stars) falls in the building category, indicating that this universal robust representation has captured the land-use representation of buildings. Other representations are scattered around the nonbuilding category since there are multiple land-use conditions in those areas. After alignment, the robust representations (red crosses) move closer to the building category (light blue rhomboid) of the specific image, demonstrating the effectiveness of the alignment operation. Meanwhile, the robust representations of the nonbuilding category remain scattered in the nonbuilding area after passing through the alignment

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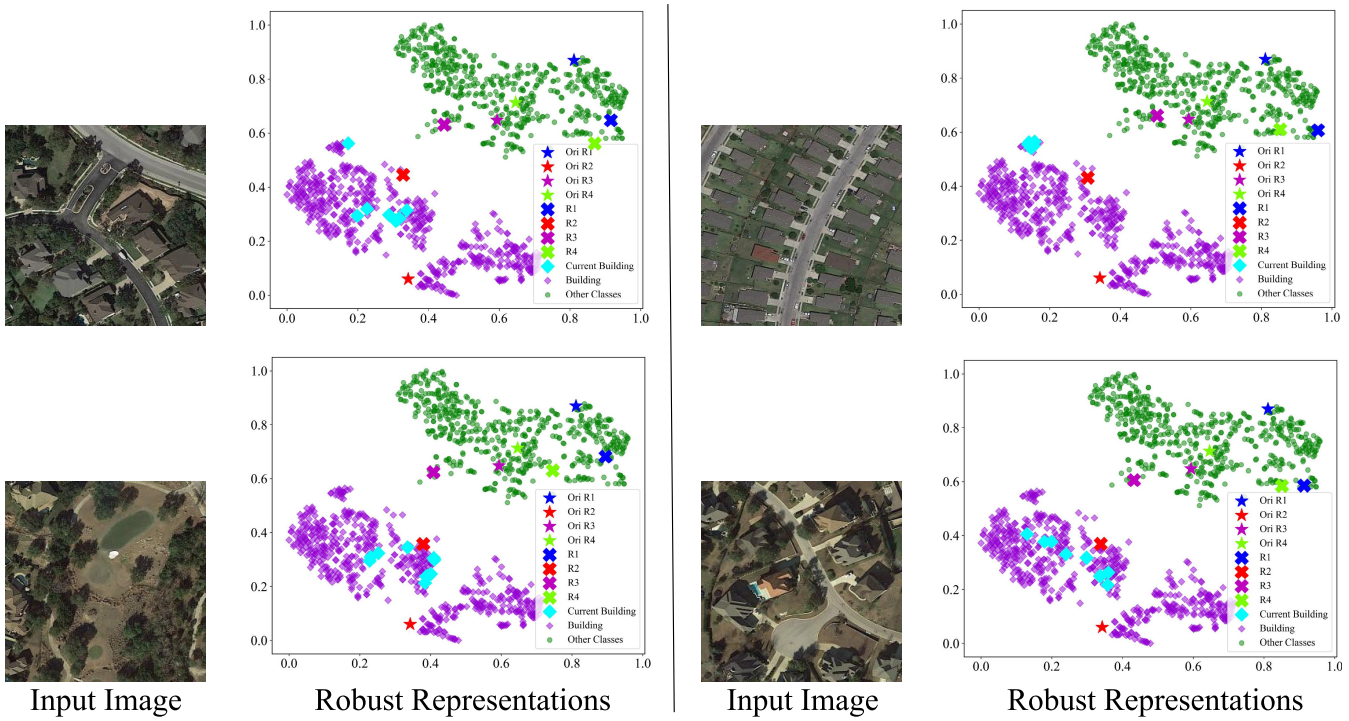


Fig. 9. Visualization of the four robust representations in LEVIR-CD before and after alignment. In the legend, "Ori R1–R4" are universal robust representations before alignment. "R1–R4" are specific robust representations after alignment to the input bitemporal images. "Current building" is the feature of the building on the input bitemporal images. "Building" is the feature of building from 58 pairs of images. "Other classes" are features of areas other than buildings in 58 pairs of images. The universal robust representation of buildings (Ori R2) is located among the "building." After alignment to the input bitemporal images, the specific robust representation (R2) is located close to the "current building."

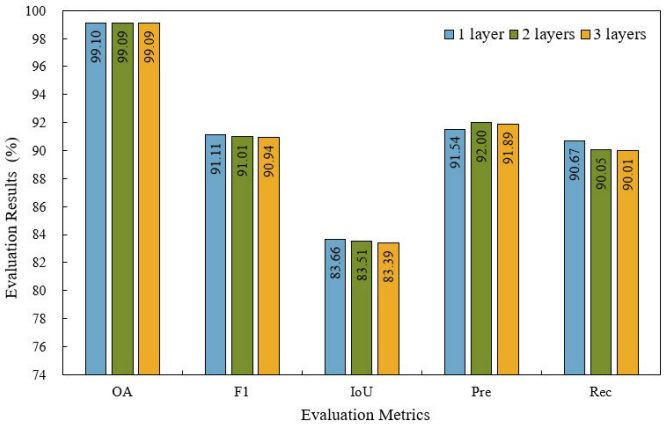


Fig. 10. Ablation study of the depth of RRGModule on the LEVIR-CD dataset.

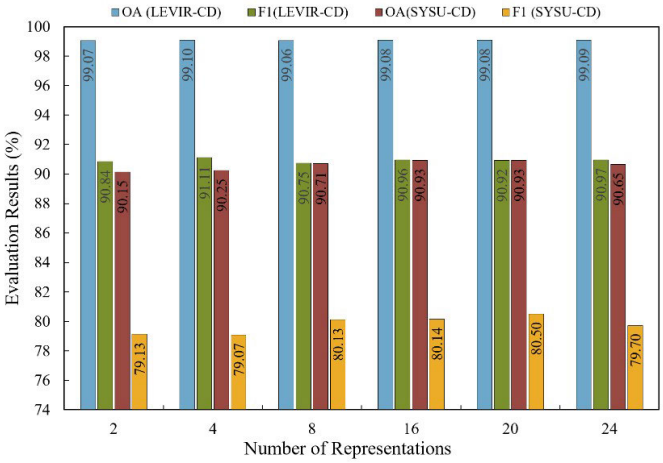


Fig. 11. Ablation study of the number of the STRobustNets on the LEVIR-CD and SYSU-CD datasets.

network, without confusion with the building representations. After alignment, the absence of confusion also proves that the representations we have learned possess spatial–temporal robustness.

3) *Numbers of ALs:*
We investigate the impact of different numbers of ALs in Fig. 10. With the increase in the number of layers, the performance decreases. This is because excessive alignment operations can reduce the discriminability of the representations, leading to a decrease in classification accuracy. Therefore, we set the number of ALs of RRGModule to 1 in the final STRobustNet.

4) *Numbers of representations:*
We explore the impact of different numbers of representations on various datasets. We perform experiments on the LEVIR-CD and SYSU-CD datasets, varying the number of representations as 2, 4, 8, 16, 20, and 24 for each dataset. The IoU loss weight α is set to 0.5 for these two datasets. The results are summarized in Fig. 11. It can be observed that on the LEVIR-CD and SYSU-CD datasets, STRobustNet achieves optimal performance with 4 and 20 representations, respectively. This is due to the fact that the number of interested land-use classes

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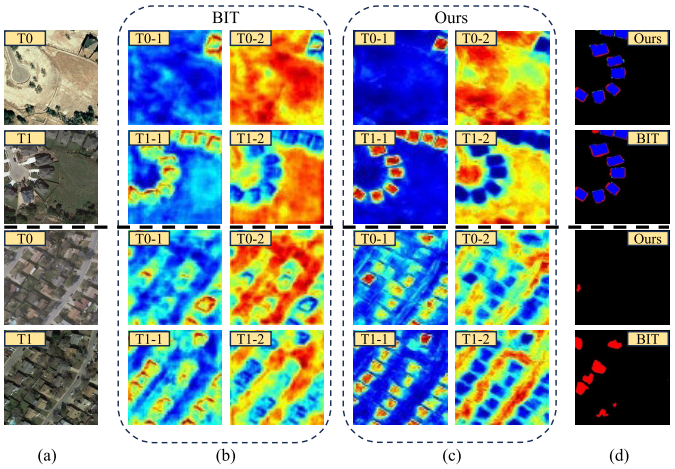


Fig. 12. Comparison between BIT and Our method. (a) Bitemporal images. (b) Visualization of BIT's regions of interest. (c) Visualization of our regions of interest. (b) and (c) Both are obtained by calculating the dot product between each representation of BIT/STRobustNet and the pixel-level features extracted from bitemporal images. (d) Change maps of BIT and ours, with **true positives** depicted in **blue**, **true negatives** in **black**, **false positives** in **red**, and **false negatives** in **green**.

varies across different datasets. Compared to LEVIR-CD, SYSU-CD has a more complex and diverse set of interested land-use classes.

5) For a more intuitive demonstration of the effectiveness of our proposed STRobustNets, we visualize the feature distribution of different regions of interest of STRobustNet and BIT. Utilizing STRobustNets for classifying bitemporal features (as shown in Fig. 12) yields results that indicate the distribution of regions of interest. In simple scenes like the first two rows of Fig. 12, although both BIT and our method can identify the regions of interest, our method provides a clearer delineation of the interested areas compared to BIT. Therefore, our detection results exhibit finer details at the boundaries. In complex scenes like the last two rows of Fig. 12, the presence of similar features between buildings and the background can lead to feature confusion when using methods such as BIT that employs self-attention mechanisms for aggregating bitemporal features. As a result, the regions of interest obtained by BIT integrate the buildings with their surrounding environment in the last two rows of Fig. 12(b), causing false detection. In contrast, our method directly extracts STRobustNets, avoiding the aforementioned issue.

Fig. 13 further visualizes the classification representations, which are obtained by activating the bitemporal features with four robust representations. It can be seen that the building and nonbuilding categories are activated in different channels. Moreover, images without buildings show no activation in the building-related channels (first group). For images with no changes, the activation patterns are consistent in the bitemporal images across all four channels (third group). This demonstrates that the designed method can obtain spatial–temporally consistent classification representations, leading to accurate CD results. It also validates the correctness of the learned STRobustNets.

2) Impact of DRModule: To further enhance the details in the change map, we introduce the DRModule. As shown in

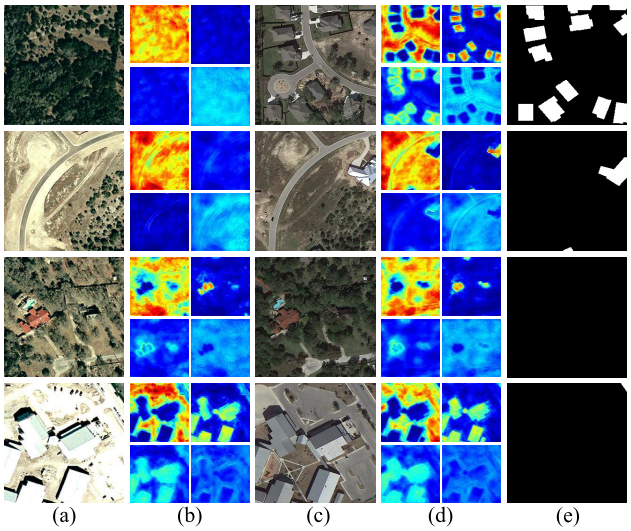


Fig. 13. Visualization of the classification representations in the LEVIR-CD dataset, where the number of the robust representations of land-use classes of interest is 4. The classification representations are derived by calculating the dot product between each robust representation and the pixel-level features extracted from bitemporal images. (a) T0 image. (b) Classification representations of T0. (c) T1 image. (d) Classification representations of T1. (e) Ground-truth change map.

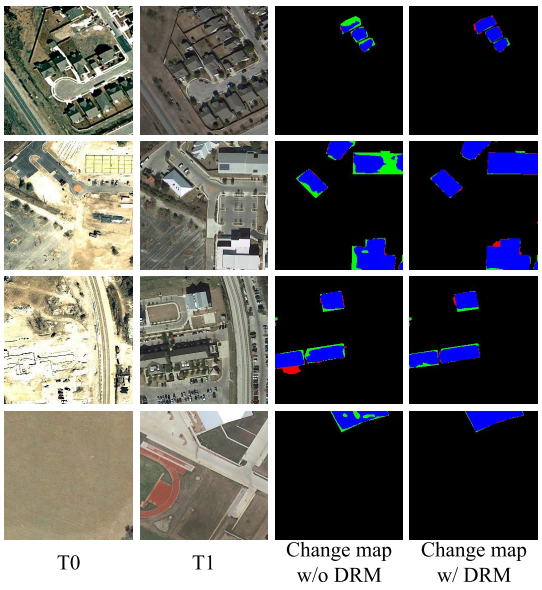


Fig. 14. Visualization of the DRModule's (DRM) effectiveness. The color scheme employed for visualization is given as follows: **blue** represents **true positives**, **black** represents **true negatives**, **red** represents **false positives**, and **green** represents **false negatives**.

Table IV, the DRModule raises the F1-Score from 89.92% to 91.11%. This is because there are certain details in the bitemporal images of the original resolution that get overlooked during the downsampling process, and these details are crucial for enhancing the accuracy of the change map. We also visualize the detection results before and after the DRModule. As shown in Fig. 14, after applying the DRModule, the occurrence of false positive and false negative areas at object boundaries is significantly reduced. This further validates the effectiveness of our proposed DRModule.

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TABLE V

IMPACT OF DIFFERENT LOSS FUNCTIONS ON LEVIR-CD. "BASELINE" REFERS TO THE MODEL WITHOUT DRMODULE AND TRAINED WITH IOU LOSS AND VANILLA CROSS-ENTROPY LOSS. "DRM:" DRMODULE. "E:" EDGE-ENHANCED CROSS-ENTROPY LOSS. "S:" SMOOTHNESS LOSS

Loss	OA	F1	IoU	Pre	Rec
Baseline	98.98	89.83	81.53	91.07	88.62
w/ DRM	99.07	90.81	83.16	91.40	90.23
w/ DRM+E	99.07	90.82	83.18	91.14	90.50
w/ DRM+E+S	99.10	91.11	83.66	91.54	90.67

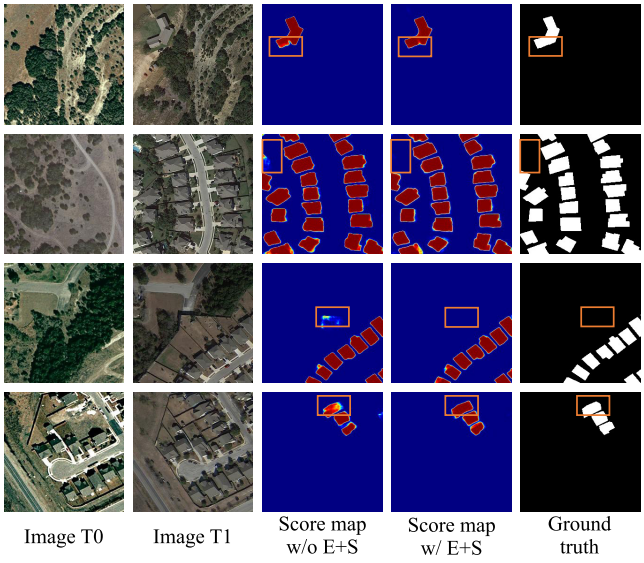


Fig. 15. Impact of edge-enhanced cross-entropy loss (E) and smoothness loss (S). The "score map" represents the probability of change occurrence. Red indicates a probability value closer to 1, suggesting a high likelihood of change, while blue indicates a probability value closer to 0, signifying a low likelihood of change. Colors in between red and blue represent areas where the network struggles to make definitive predictions, i.e., there is ambiguity in the network's predictions of change for these regions.

3) Impact of Different Loss Functions: We conduct ablation experiments to better understand the roles of edge-enhanced cross-entropy loss (E) and smoothness loss (S). As shown in Table V, although the inclusion of DRModule significantly enhances the "Baseline," it may introduce redundant information. Therefore, when "E" is incorporated into the training, both IoU and recall are further improved, indicating an expansion of the true positive change regions. However, a slight decrease in precision is observed, likely due to reduced constraints within objects, which introduced noise. When both "E" and "S" are applied simultaneously, all metrics show substantial improvements. This demonstrates that the combination of these losses effectively addresses the imbalance between object boundaries and interior regions. We further visualize the score maps with or without "E + S" in Fig. 15. In the absence of the "E + S" constraints, the DRModule fails to adequately refine details at the edges of changed areas, resulting in ambiguous score predictions, as highlighted within the orange boxes. When the "E + S" is applied, the scores in these ambiguous regions become more definitive, as indicated by the intensification of red (changed areas) or blue (unchanged areas). This observation demonstrates that the "E + S" enhances the DRModule's ability to focus on detail refinement.

V. CONCLUSION

In this article, we propose a novel CD framework called STRobustNet. Unlike previous approaches that directly aggregate context on bitemporal features to address spatial–temporal inconsistency, STRobustNet addresses the issue by detecting changes in the spatial–temporal consistent classification representations. To obtain the spatial–temporal consistent classification representations, STRobustNet extracts a set of STRobustNets to classify the bitemporal features. The acquisition of robust representations considers both universal spatial–temporal features from the large-scale bitemporal scene and specific spatial–temporal knowledge from the current bitemporal scene, making the detection more robust to appearance inconsistencies caused by temporal and spatial variations of the same land-use class. Furthermore, STRobustNet avoids the need for self-attention computation across the entire feature map, thereby mitigating feature confusion issues and significantly improving the efficiency compared with existing transformer-based methods.

However, STRobustNet still has its limitations. It is designed under the assumption of perfect registration between bitemporal images. Consequently, STRobustNet's performance is compromised when there is a misalignment between bitemporal images. Since STRobustNet relies on pixel classification where neighboring pixels are classified into the same category, misalignments within the interior of objects in the two images do not affect CD. The impact is primarily concentrated on the boundary regions of objects, as the two sides of a boundary are classified into different object categories. When the two images are not aligned, issues such as missed detections or false positives may arise in the boundary regions of objects. In our future work, we will investigate methods to determine whether changes have occurred based on the neighboring regions of corresponding pixels in the classification representations, enhancing the robustness of misalignment between bitemporal images.

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